

The Landau-Selberg-Delange method for Dirichlet L -functions, and applications

Akash Singha Roy
Charles University, Prague

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Extension? Assume that $\sum_n a_n/n^s = \left(\prod_{\chi \bmod q} L(s, \chi)^{\alpha_\chi} \right) \cdot G(s)$ for all $\operatorname{Re}(s) > 1$, where $\{\alpha_\chi\}_\chi \subset \mathbb{C}$ and $G(s)$ is “well-behaved”.

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- Give **as precise asymptotic series** estimating $\sum_{n \leq x} a_n$, **uniformly in q in a wide range** (wider than from previous literature).

One of the main results

Theorem 1 (S.R., outline).

Fix $\nu \in \mathbb{R}^+$. Assume $\sum_n a_n/n^s = \left(\prod_{\chi \bmod q} L(s\nu, \chi)^{\alpha_\chi}\right) \cdot G(s)$ for all s with $\operatorname{Re}(s) > 1/\nu$, where $\{\alpha_\chi\}_\chi \subset \mathbb{C}$, and $G(s)$ is some “well-behaved” function. Then for any fixed $K_0 > 0$, we have

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uniformly in $x \geq 3$, $N \geq 0$ and $q \leq (\log x)^{K_0}$.

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Wider ranges under Landau-Siegel zeros conjecture and GRH.

Some applications

- (1) Positive integers with prime divisors restricted to residue classes:
Given $q \in \mathbb{Z}^+$ and $\mathcal{A} \subset (\mathbb{Z}/q\mathbb{Z})^\times$, estimate
 $\#\{n \leq x : p \mid n \implies p \bmod q \in \mathcal{A}\}.$

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(2) Distributions of the least invariant factor of multiplicative groups:

Writing $(\mathbb{Z}/n\mathbb{Z})^\times \cong \mathbb{Z}/\lambda_1\mathbb{Z} \oplus \mathbb{Z}/\lambda_2\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/\lambda_r\mathbb{Z}$ with $\lambda_i \in \mathbb{Z}^+$, $\lambda_1 \mid \lambda_2 \mid \cdots \mid \lambda_r$, let $\lambda_1(n) := \lambda_1$. Estimate $\#\{n \leq x : \lambda_1(n) = d\}.$

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- Chang–Martin (2020): Fixed $d \in \mathbb{Z}^+$.
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 - Least primary factors** of unit groups:
Extends Martin–Nguyen (2024).

More applications

(3) Sathe–Selberg Theorem in arithmetic progressions:

- Sathe–Selberg: Local distributions of the functions $\omega(n) = \sum_{p|n} 1$ and $\Omega(n) = \sum_{p|n} v_p(n)$.

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$$\omega_a(n) := \sum_{\substack{p|n \\ p \equiv a \pmod{q}}} 1 \quad \text{and} \quad \Omega_a(n) := \sum_{\substack{p|n \\ p \equiv a \pmod{q}}} v_p(n).$$

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More applications

- (5) **Residue-class distribution of multiplicative functions:** Given multiplicative functions $f_1, \dots, f_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$, estimate $\#\{n \leq x : (\forall i) f_i(n) \equiv a_i \pmod{q}\}$ for units $a_i \pmod{q}$.

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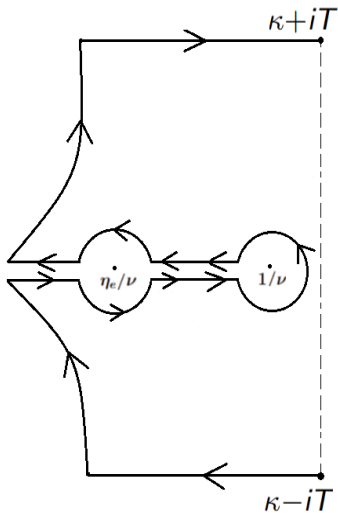
Aside (Ingredients): “Mixing” in $(\mathbb{Z}/q\mathbb{Z})^\times$, detected via: anatomy of integers, character sum machinery, linear algebra over rings, arithmetic geometry, algebraic geometry, Theorem 1.

Main ideas behind Theorem 1: Summarized

Let $\mathcal{F}(s) = \prod_{\chi \bmod q} L(s\nu, \chi)^{\alpha_\chi}$. Then

$$\sum_{n \leq x} a_n \approx \frac{1}{2\pi i} \int_{\kappa-iT}^{\kappa+iT} \frac{\mathcal{F}(s)G(s)x^s}{s} ds$$

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Perron error for a general $\nu > 0$.)



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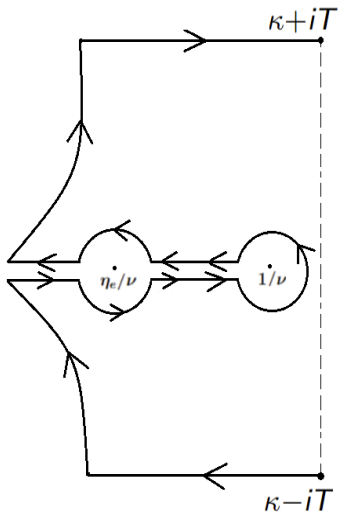
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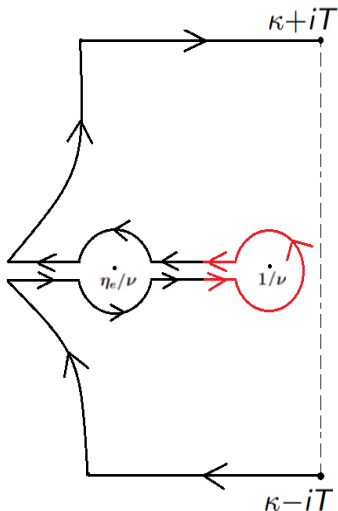
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$$\begin{aligned} &\int_{\text{Red part}} \frac{\mathcal{F}(s)G(s)x^s}{s} ds \\ &\longrightarrow \text{main term, secondary term, } \dots \end{aligned}$$

Need: Bound on $|\mathcal{F}(s)|$ on the rest of Γ .



Bounding $\mathcal{F}(s) = \prod_{\chi \bmod q} L(s\nu, \chi)^{\alpha_\chi}$ on the rest of Γ

Idea 1. $\lambda_q := 1 + \max_{a \bmod q} \left| \sum_{\chi \bmod q} \alpha_\chi \cdot \chi(a) \right|$

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→ Split into regions + (log-free) zero-density estimates
+ various manipulations with Stieltjes integrals.

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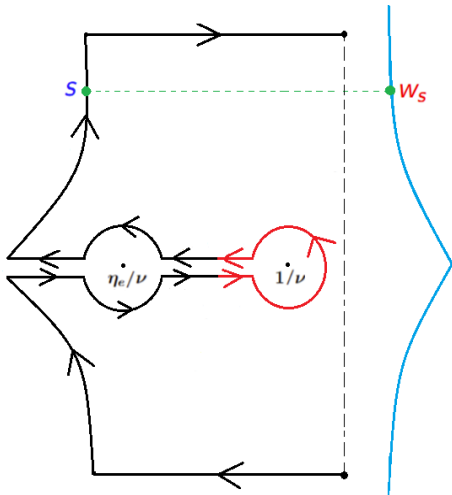
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Idea 3. Deduce bound on $\mathcal{F}(s)$ via suitable “auxiliary functions”.

$$\begin{aligned} \mathcal{H}(s) \\ = \mathcal{F}(s) (s - 1/\nu)^{\alpha_{\chi_0}} (s - \eta_e/\nu)^{-\alpha_{\chi_e}} \end{aligned}$$



Bounding $\mathcal{F}(s) = \prod_{\chi \bmod q} L(s\nu, \chi)^{\alpha_\chi}$ on the rest of Γ

Idea 1.

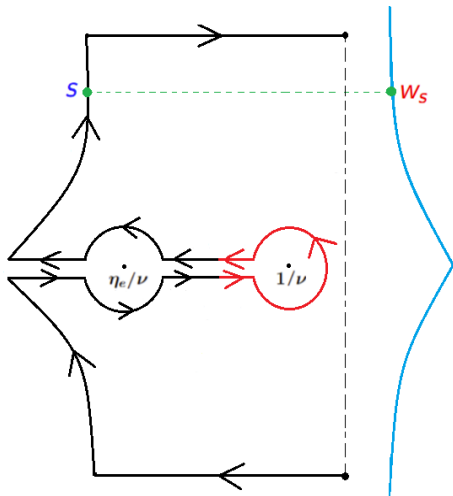
$$\lambda_q := 1 + \max_{a \bmod q} \left| \sum_{\chi \bmod q} \alpha_\chi \chi(a) \right|.$$

Idea 2. Bound $\mathcal{F}'(s)/\mathcal{F}(s)$.

Idea 3. Deduce bound on $\mathcal{F}(s)$ via suitable “auxiliary functions”.

$$\begin{aligned} \mathcal{H}(s) \\ = \mathcal{F}(s) (s - 1/\nu)^{\alpha_{\chi_0}} (s - \eta_e/\nu)^{-\alpha_{\chi_e}} \end{aligned}$$

- Relate values of $\log \mathcal{F}(s)$ on Γ with values of $\log \mathcal{F}(w_s)$.
- **Dirichlet series** for $\log \mathcal{F}(w_s)$.



Thank you for your attention!

Email: akash01s.roy@gmail.com